Zachary R. Zeisler, Ph.D

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CURRENT POSITION

Université de Montréal, Mila — Montréal, QC
Postdoctoral Fellow — October 20, 2026 - present
Advisor: Matthew Perich, PhD
Project: Cognitive-motor integration during primate reaching and grasping
Techniques: NHP behavior training, analysis, computational modeling; high-density electrophysiology recordings and analysis; large animal surgery (implants, viral injections, perfusions)

EDUCATION

Icahn School of Medicine at Mount Sinai — New York, NY
Postdoctoral Fellow — April 1, 2024 – October 16, 2026
Advisor: Peter Rudebeck, D.Phil
Project: Circuit mechanisms of reinforcement learning in macaques

Icahn School of Medicine at Mount Sinai — New York, NY
Ph.D Neuroscience — February 27, 2024
Advisor: Peter Rudebeck, D.Phil
Thesis: A comparative analysis of amygdala anatomy and physiology in mice, macaques, and man

Mercer University — Macon, GA
B.S. Neuroscience, B.A. Latin; *summa cum laude* — May, 2019
Advisors: Katherine Northcutt, Ph.D & Achim Kopp, Ph.D

PUBLICATIONS

London, L., Love, M., **Zeisler, Z.R.**, Rudebeck, P.H., Stoll, F.M. 2026
Dataset of cortical and subcortical single neuron activity during value-based tasks in macaque monkey. *Scientific Data.*

Zeisler, Z.R., Rudebeck, P.H. 2025
Evolutionary road to primate prefrontal cortex, in *The Evolution of Nervous Systems*, 3rd edition, Volume 3, *The Evolution of Nervous Systems in Nonhuman Primates*, E. A. Murray and S. P. Wise, eds.

Zeisler, Z.R., Love, M., Rutishauser, U., Stoll., F.M., Rudebeck, P.H. 2025
Consistent hierarchies of single-neuron timescales in mice, macaques, and humans. *Journal of Neuroscience. 45 (19)* e2155242025.

Zeisler, Z.R., Heslin, K.A., Stoll, F.M., Hof, P.R., Clem, R.L., Rudebeck, P.H. 2024
Comparative basolateral amygdala connectomics reveals dissociable single-neuron projection patterns to frontal cortex in macaques and mice. *Current Biology 34:14,* 3249-3257.E3.

Zeisler, Z.R., London, L., Janssen, W.G., Fredericks, J.M., Elorette, C., Fujimoto, A., Zhan, H., Russ, B.E., Hof, P.R., Clem, R.L., Stoll, F.M., Rudebeck, P.H. Single basolateral amygdala neurons in macaques exhibit distinct connectional motifs with frontal cortex. *Neuron 111,* 3307-3320. 2023

INVITED TALKS

New York Memory Hub Conference September 26, 2025, New York, NY
Zeisler, Z.R., Stoll, F.M., London, L., Folloni, D., Rudebeck, P.H. The neural circuits and mechanisms of short-term memory during reward learning.

Janelia Mechanistic Cognitive Neuroscience Junior Scientist Workshop March 2-7, 2025, Ashburn, VA
Zeisler, Z.R., Folloni, D., Stoll., F.M., Rudebeck, P.H. Circuit mechanisms of bottom-up influences on reversal learning in macaques.

Friedman Brain Institute Annual Retreat May 12, 2023, New York, NY
Zeisler, Z.R., Heslin, K., Clem, R., Rudebeck, P.H. A comparative analysis of amygdala neuron branching in macaques and mice.

Neuro RECEPTR, Rutgers University, Brain Health Institute April 27-28, 2023, New Brunswick, NJ
Zeisler, Z.R., London, L., Janssen, W.G., Fredericks, J.M., Elorette, C., Fujimoto, A., Zhan, H., Russ, B.E., Hof, P.R., Clem, R.L., Stoll, F.M., Rudebeck, P.H. High-throughput sequencing of macaque basolateral amygdala projections reveals dissociable connectional motifs with frontal cortex.

UCLA Science Outreach Symposium November 10, 2022, Los Angeles, CA
Zeisler, Z.R., Loo, H. Project SHORT: Diversifying STEM through one-on-one mentorship.

AWARDS AND HONORS

Cajal Club: Krieg Cortical Scholar 2025

POSTER PRESENTATIONS

Computational and Systems Neuroscience (COSYNE) March 27-April 1, 2025, Montreal, Quebec
Zeisler, Z.R., Stoll., F.M., Folloni, D., Perich, M.G., Rudebeck, P.H. Neural circuit mechanisms of bottom-up reward learning.

From Neuroscience to Artificially Intelligent Systems Sept. 28-Oct. 2, 2024, Cold Spring Harbour, NY
Zeisler, Z.R., Stoll, F.M, Rudebeck, P.H. Neural dynamics supporting affective reward learning

Winter Conference on Brain Research Jan. 27-Feb. 2, 2024, Breckenridge, CO
Zeisler, Z.R., Heslin, K., Clem, R.L., Rudebeck, P.H. The comparative organization of single amygdala neuron projection patterns to frontal cortex and striatum in macaques and mice

Society for Neuroscience November 11-15, 2023, Washington, DC
Zeisler, Z.R., Minxha, J., Rutishauser, U., Stoll, F.M., Rudebeck, P.H. Conserved hierarchies of single-unit timescales in frontal and limbic regions in mice, macaques, and humans.

Society for Neuroscience November 11-15, 2023, Washington, DC
Zeisler, Z.R., Heslin, K., Clem, R.L., Rudebeck, P.H. The comparative organization of single amygdala neuron projection patterns to frontal cortex and striatum in macaques and mice

Society for Neuroscience November 11-17, 2022, San Diego, CA
Zeisler, Z.R., London, L., Janssen, W.G., Fredericks, J.M., Elorette, C., Fujimoto, A., Zhan, H., Russ, B.E., Hof, P.R., Clem, R.L., Stoll, F.M., Rudebeck, P.H. High-throughput sequencing of macaque basolateral amygdala projections reveals dissociable connectional motifs with frontal cortex.

Gordon Research Conference – Frontal Cortex August 8-12, 2022, Ventura, CA
Zeisler, Z.R., Fredericks, J.M., Janssen, W.G., Stoll, F.M., Clem, R.L., Rudebeck, P.H. High-throughput sequencing of macaque basolateral amygdala projections reveals dissociable connectional motifs with frontal cortex.

Friedman Brain Institute Neuroscience Retreat May 13, 2022, New York, NY
Zeisler, Z.R., Fredericks, J.M., Janssen, W.G., Clem, R.L., Rudebeck, P.H. Single neurons in macaque basolateral amygdala exhibit distinct patterns of collateral projections.

International Congress of Neuroendocrinology July 15-18, 2018, Toronto, Ontario
Zeisler, Z.R., Smith, S., Northcutt, K.N. Sex-specific modulation of juvenile midbrain dopamine expression by perinatal serotonin.

Society for Neuroscience November 11-15, 2017, Washington, DC
Zeisler, Z.R., Smith, S., Northcutt, K.N. Perinatal hyperserotonemia influences dopamine expression in the midbrain.

ORAL PRESENTATIONS

Society for Neuroscience November 15-19, 2025, San Diego, CA
Zeisler, Z.R., Stoll, F.M., London, L., Folloni, D., Perich, M.G., Rudebeck, P.H. Neural circuit mechanisms of bottom-up reward learning. Nanosymposium.

Friedman Brain Institute Cells, Circuits, and Behavior WIP December 5, 2024, New York, NY
Zeisler, Z.R., Folloni, D., Stoll, F.M., Rudebeck, P.H. Neural circuit mechanisms of bottom-up reinforcement learning.

Friedman Brain Institute Cells, Circuits, and Behavior WIP December 1, 2022, New York, NY
Zeisler, Z.R., London, L., Janssen, W.G., Fredericks, J.M., Elorette, C., Fujimoto, A., Zhan, H., Russ, B.E., Hof, P.R., Clem, R.L., Stoll, F.M., Rudebeck, P.H. High-throughput sequencing of macaque basolateral amygdala projections reveals dissociable connectional motifs with frontal cortex.

Center for Neurotechnology and Behavior Seminar Series March 15, 2022, New York, NY
Zeisler Z., Stoll, F.M., Love, M., Rudebeck, P.H. Intrinsic timescales reflect neuroanatomical differences across species.

Friedman Brain Institute Cells and Circuits WIP January 14, 2022, New York, NY
Zeisler, Z.R., Fredericks, J.M., Janssen, W.G., Stoll, F.M., Clem, R.L., Rudebeck, P.H. The Primate Amygdala: Projections at single-neuron resolution.

PEER REVIEW ACTIVITY

Cell Reports, Communications Biology, Journal of Neuroscience

OTHER EDUCATION

Flatiron Institute, Center for Computational Neuroscience Software Workshop February 4-6, 2025
Training in use of open-source Python packages *Pynapple* and *NeMos* to analyze, model, and visualize neural data.

Machine Learning for Large-Scale Neuroscience October 9, 2021
Workshop attended as part of 2021 CRCNS Meeting.

NeuroMatch Academy July, 2020 – August, 2020
3-week intensive computational neuroscience training course: Python programming, machine learning, big data; group project working with Human Connectome Project fMRI data

OUTREACH ACTIVITIES

Project SHORT – Director of Outreach July, 2020 – August, 2023
Assisted with organization and recruitment of students from underrepresented communities to receive *pro bono* admissions consulting for graduate or medical school. Oversaw redesign of website.

MiNDS – Mentoring in Neuroscience Discovery at Sinai August, 2019 – August, 2023
Annual Brain Fair during Brain Awareness Week; monthly classes at local middle school; other STEM advocacy activities across NYC

TEACHING EXPERIENCE

Icahn School of Medicine New York, NY
Teaching Assistant, Behavioral and Cognitive Neuroscience (for PhD students) Spring 2021, 2022
CEYE – Sherman Scholars program August, 2020, 2021, 2022
MiNDS – Intro neuroscience course for middle school students August, 2019 – August, 2021

Mercer University Macon, GA
Lab Teaching Assistant, Intro to Environmental Science Fall 2018
Lab Teaching Assistant, Intro to Biology I Spring 2017, Spring 2018
Lab Teaching Assistant, Intro to Biology II Fall 2017

INSTITUTIONAL SERVICE

Icahn School of Medicine New York, NY
Neuroscience Graduate Program: Interviews, Panels, Socials Spring 2020, 2021, 2022, 2023, 2024